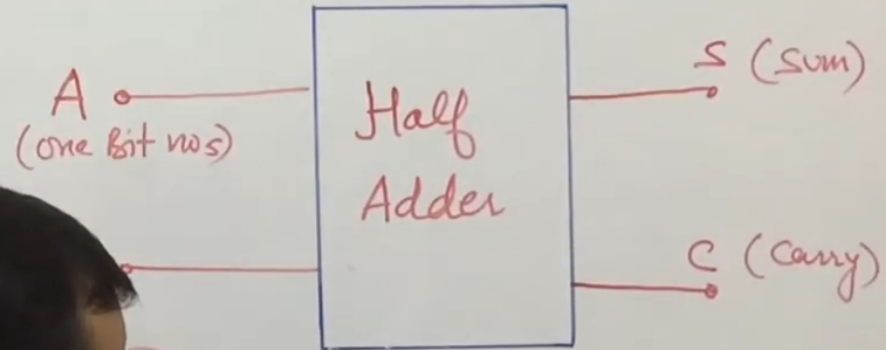


Designing of a Half Adder - One bit Addition

* A logic ckt. for the addition of two one bit numbers is called Half Adder



2^n
 $n \rightarrow$ no. of I/Os
 $2^2 = 4$

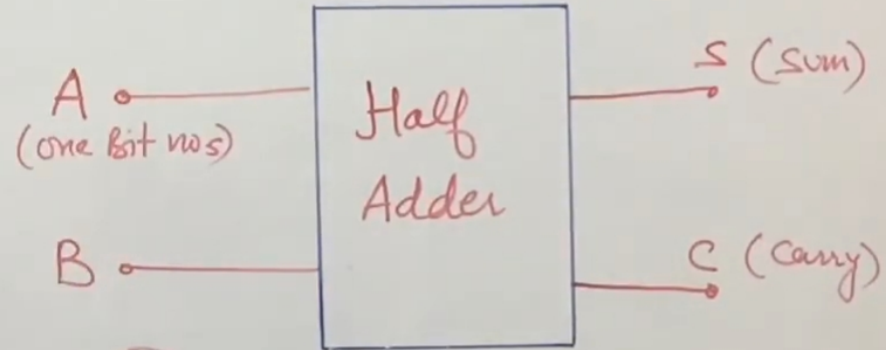
I/Ps		S
A	B	
0	0	0
0	1	1
1	0	1
1	1	0

$$S = A\bar{B} + \bar{A}B = A \oplus B$$
$$C = A \cdot B$$

Designing of a Half Adder

* A logic ckt. for the addition of two one bit numbers is called Half Adder.

I/Ps		O/Ps	
A	B	$\check{S}$	$\check{C}$
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1



		S	
A	B	0	1
0	0	0	1
0	1	1	0
1	0	1	0
1	1	0	1

$$S = A\bar{B} + \bar{A}B = \underline{A \oplus B}$$

$$C = \underline{A \cdot B}$$

